

Luca MORELLI

Computer scientist - University of Salerno

in [linkedin.com/in/luca-morelli-bn2000/](https://www.linkedin.com/in/luca-morelli-bn2000/) github.com/LucoMoro
☎ +39 380 10 83 212 @ morelli.2000luca@gmail.com
📍 C.da Botticella 79, Campolattaro (BN), Italy
📅 Born in 14 October 2000 (24 years old) in Benevento, Italy



EDUCATION

2023–Current	Master's Degree in Software Engineering and IT Management, <i>University of Salerno</i> Thesis title : CRANE – Code Review AI Network Engine Expected graduation : July 2025 Weighted average : 29.81 / 30 Completed exams : 15 / 15
2019–2023	Bachelor's Degree in Computer Science, <i>University of Salerno</i> Thesis title : DeepGravity Tool, the DeepGravity user interface Final grade : 104 / 110

SKILLS

Programming Languages	Java, Python, C, C++
Software Security	Vulnerability awareness (MITRE CWE Top 25, OWASP Top 10), static and dynamic code analysis (SonarCloud, Semgrep)
AI & Machine Learning	LLM usage (OpenAI API, Hugging Face API), Python ML libraries (scikit-learn)
Prompt Engineering Patterns	Zero-shot, Few-shot, Chain-of-Thought, Flipped-Interaction, Persona prompting
Software Engineering	Testing, CI/CD, version control (Git), design patterns

EXPERIENCES

26 Feb 2025 31 May 2025	Erasmus+ Traineeship, JHERONIMUS ACADEMY OF DATA SCIENCE (JADS), Netherlands Currently working on a security-tailored version of CRANE, aimed at detecting and repairing code vulnerabilities. The project involves (i) building a collaborative pipeline of security-aware LLM agents powered by prompt engineering and custom RAGs, (ii) conducting experiments to evaluate its real-world effectiveness. Secure Software Development Code Vulnerability Detection LLMs Prompt Engineering AI-Driven Security
09 May 2023 07 Nov 2023	Tutoring Activity – Algorithm Design, UNIVERSITY OF SALERNO (UNISA), Italy Provided academic support to over 200 students in the "Algorithm Design" course. Assisted during both theoretical lectures and practical sessions and led pre-exam workshops and exercises. Algorithm Design Complexity Analysis Mentoring Divide-and-Conquer Greedy Algorithms Academic Support

LANGUAGES

- > Italian (Native)
- > English (B2)

+ STRENGTHS

- > Intellectually curious
- > Strong collaborative mindset
- > Capable of performing in high stress environments

PUBLICATIONS

[C1] **Luca Morelli**, Gilberto Recupito, Antonio Della Porta, Giammaria Giordano, Fabio Palomba, "When Personality Is not Enough : A Replication Study on Personality-Based Team Performance Prediction in Educational Contexts", under submission at the 51st Euromicro Conference on Software Engineering and Advanced Applications (SEAA 2025), Salerno, Italy.

This study investigates the adaptability of an existing rule-based model for software development team composition, designed to predict team effectiveness based on members' personality traits, in a new context.

[Team Composition](#) [Personality Traits](#) [Software Engineering Management](#) [Empirical Validation](#)

PROJETS

CRANE – CODE REVIEW AI NETWORK ENGINE (MASTER THESIS PROJECT)

NOV 2024 – PRESENT

github.com/LucoMoro/CRANE [Project on GitHub](#)

CRANE is a research-driven system exploring how Large Language Models (LLMs) can enhance the code review process via multi-agent collaboration. The system models a team of specialized LLM agents that engage in structured dialogue to analyze, critique and resolve code review comments. The workflow is designed to produce high-quality, actionable feedback to assist both humans and LLMs in secure and effective code revision.

[LLMs](#) [Prompt Engineering](#) [Multi-Agent Systems](#) [Code Quality](#) [AI for Software Engineering](#)